

Bank fintech, liquidity creation, and risk-taking: Evidence from China

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ABSTRACT

The impact of fintech on banking industry has garnered significant academic attention. While existing research has primarily focused on the influence of fintech on bank performance, little attention has been paid to its impact on liquidity creation. This study examines the relationship between fintech and bank liquidity creation and risk-taking using bank-level data for Chinese listed banks from 2007 to 2021. Using text analysis, we develop a novel fintech index to measure the fintech development. Results show that fintech development significantly increases bank liquidity creation. This effect is particularly pronounced for banks with larger size, higher ownership concentration, and greater market power. The impact of fintech on liquidity creation primarily stems from changes in banks' credit structure. Additionally, fintech stimulates bank risk-taking through the liquidity creation channel, highlighting the need for rigorous risk management. This study provides practitioners and regulators with new insights into the impact of fintech on banking.

1. Introduction

In the era of innovation-driven development and the ongoing deepening of financial supply-side reforms, China's economy and society are experiencing a profound transformation propelled by a new wave of technological revolution driven by information technology. The acceleration of digitalization is reshaping the models of economic development and social governance. The limitations of the traditional development model are becoming more evident, necessitating the exploration of a high-quality economic growth model that aligns with the demands of this new technological revolution. By 2021, the size of the digital economy in China had expanded to 45.5 trillion yuan, representing approximately 39.8 percent of the country's GDP,¹ bringing significant opportunities to the reform of the financial industry. According to patent data from the World Intellectual Property Organization (WIPO), more than 7000 companies around the world applied for 85,400 fintech patents, and 19,400 were authorized from 2017 to the first half of 2021. Among these, China ranks first and second in the world in terms of fintech patent applications and grants, with 37,500 and 3720 respectively. Eight Chinese companies are on the list of the top 10 companies in the world in terms of the number of patent applications,

indicating that China's fintech development has taken the lead in the world. The integration of fintech plays a crucial role in China's endeavors to deepen supply-side reforms, enhance the financial sector's capacity to support the real economy, and prevent and mitigate financial risks. As the mainstay of China's financial system, commercial banks are the main providers of indirect financing and an important carrier of "technology-enabled finance". Especially since 2020, the continuous spread of COVID-19 has slowed down the economic recovery, and most enterprises were in urgent need of capital turnover, debt repayment, financing expansion, and resumption of work and production, which required liquidity support from commercial banks and relies more on online financial services (Fu and Mishra, 2022). Therefore, commercial banks have embraced the wave of fintech and accelerated digital transformation (Kotarba, 2016), which is an inevitable choice for commercial banks to improve quality and efficiency and effectively support the real economy.

Financial intermediation theory suggests that commercial banks' core functions are liquidity creation and risk transformation (Diamond and Dybvig, 1983; Bhattacharya and Thakor, 1993). The existing literature on the influencing factors of liquidity creation mainly focuses on the internal and external factors of banks. The external factors affecting

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¹ The data is taken from the white paper entitled "Jointly Building a Community of Shared Future in Cyberspace" published by the State Council Information Office of the People's Republic of China.

bank liquidity creation mainly include monetary policy (Bernanke and Blinder, 1992), deposit insurance system (Fungáčová et al., 2017), regulatory intensity (Distinguin et al., 2013; Berger et al., 2016), and market competition (Diamond and Rajan, 2001; Black and Strahan, 2002), among others. Some studies also explore the association between bank liquidity creation and internal factors, such as bank capital (Berger and Bouwman, 2009), diversification (Hou et al., 2018), managerial ability (Andreou et al., 2016), internal governance (Díaz and Huang, 2017), and market power (Berger et al., 2022), etc.

The development of fintech provides unprecedented opportunities for the transformation and innovation of commercial banks. The traditional bank credit model pays more attention to the bank-enterprise relationship, corporate financial status and collateral. The development of lending technology has lowered the entry barrier (Vučinić, 2020). Banks utilize fintech to carry out data integration and intelligent analysis, include long-tail customers such as small and medium-sized enterprises that have long been excluded from the traditional financial system into the category of credit, and improve the inclusiveness of financial services (Financial Stability Board, 2017; Huang et al., 2020). However, the development of fintech in banks also brings challenges to their risk prevention, because it leads to the sinking of bank customers' credit ratings, improves banks' leverage ratio, and increases bank risk-taking (Financial Stability Board, 2017). Does fintech increase bank liquidity creation? What role does liquidity creation play in the influence of fintech on bank risk-taking? Investigating the above issues is of great importance and practical value to enhance the financial sector's capacity to support the real economy and maintain financial stability in the long run.

In this study, we empirically test the impact of bank fintech on liquidity creation and risk-taking using data from listed banks in China. Our study can provide micro empirical evidence for the effect of fintech enabling banks in the context of digital transformation. The main contributions of our work include three aspects:

First, we construct a new bank fintech index using text analysis. Previous studies on fintech measurement mostly made use of the scale of third-party payments and other industries, or amount of financing by fintech companies to measure the development of fintech (Hou et al., 2016; Lee et al., 2021), which measures fintech development mainly in the time series dimension rather than the cross-sectional dimension and does not reflect well the individual characteristics of banks. Some related studies employ the Chinese Digital Finance Index developed by Peking University (Qiu et al., 2018), but this annual measure focuses on the development of fintech at the provincial level. However, our new fintech index with quarterly frequency can well reflect the continuous development of fintech at the bank level. Some studies that construct a banking fintech index based on the number of Baidu search news items (Cheng and Qu, 2020), and the underlying keywords are more focused on technology, while the combination of digital technology and financial scenes is relatively weak. In this study, 155 keywords related to bank fintech are selected from three aspects of strategy construction, infrastructure and business application when constructing bank fintech indicators. These three aspects can more comprehensively reflect the level of development of bank fintech, as they encompass not only the underlying technology, but also the application of technology in various financial scenarios. In addition, a few studies use annual report text mining to construct bank fintech indexes. However, the annual report may suffer from the problem of selective disclosure and window dressing, which may reduce the objectivity of the index. Our bank fintech index is based on the text analysis of news from more comprehensive sources and thus tends to be more objective than those used in the literature.

Second, we extend the research on the factors affecting bank liquidity creation. This study contributes to the literature on the link between bank fintech and liquidity creation, as there are few studies focusing on this issue. In particular, a large body of literature examines the impact of fintech development on banks' external business (Hou

et al., 2016; Lee et al., 2021; Zhao et al., 2022) and finds a substitution effect of fintech, but there is little research on development driven by technology empowerment within banks. Our research provides a useful complement in this regard. Moreover, we also explore the heterogeneity that bank fintech to increase liquidity creation in terms of capital scale, market power and ownership.

Third, we identify the channel through which fintech affects bank liquidity creation. We find that fintech improves liquidity creation by expanding bank lending to long-tail customers and changing the structure of loans. In addition, we provide evidence that fintech increases bank risk-taking through liquidity creation.

The remaining sections of this paper are structured as follows: Section 2 provides an overview of the related literature and develops the hypotheses. Section 3 presents our sample and main variables. Section 4 describes our model and conducts a series of robustness tests. Section 5 examines the heterogeneity of fintech affecting bank liquidity creation under different bank characteristics. Section 6 explores the various channels through which fintech impacts bank liquidity creation and risk-taking. Finally, Section 7 provides the concluding remarks of the paper.

2. Literature review and hypothesis development

Over the past decade, the rapid advancement of fintech has reshaped modern financial services, emphasizing the capacity for financial supply, integrated data resources, and risk management driven by technology and risk awareness. Commercial banks have increasingly relied on cutting-edge technologies such as big data, blockchain, artificial intelligence, and cloud computing to drive innovation in financial products and service models. This innovation aims to reduce information asymmetry and barriers to entry, resulting in significant transformations in banks' business behavior and structure. Consequently, these changes have a profound impact on the level of liquidity creation within banks and its influence on the real economy.

First, banks use fintech to expand their customer base and achieve credit expansion. Specifically, banks rely more on manual verification in loan marketing, customer acquisition, credit qualification examination and other processes, resulting in high time costs and costs that hinder banks' credit issuance to some extent. At the same time, commercial banks lend more funds to large state-owned enterprises and customers with collateral, while micro, small, and medium customers are often ignored by commercial banks due to insufficient cash flow and collateral, and face financing constraints (Berger and Udell, 2006). In the context of enhanced supervision, intensified competition, financial disintermediation and policies such as common prosperity, commercial banks are gradually realizing the importance of micro, small, and medium customers. Fintech offers a possible solution for commercial banks to expand their long-tail customers on a large scale. Some studies show that network financing lowers the financing threshold for small and medium-sized enterprises. Banks have the capability to leverage fintech to convert "soft" information into "hard" information, thereby enhancing the convenience of information transmission (Cenni et al., 2015) and reduces the requirement for corporate collateral. Relational lending based on soft information can increase loan availability for micro, small, and medium enterprises (Carletti, 2004), help entrepreneurs cross the threshold of capital constraints (Yin et al., 2019), improve financial coverage, reduce the degree of financial exclusion of micro, small, and medium enterprises and farmers, and allocate financial resources rationally. Thus, the efficiency of financial resource utilization can be improved (Zhao et al., 2022), and the income of enterprises and households can be increased (Zhang and Naceur, 2019), thus reducing the Macmillan gap. Therefore, with the help of fintech, commercial banks have improved the bank credit business model and effectively promoted the credit supply of small and medium-sized enterprises (Sheng, 2021), thus increasing the liquidity creation at the asset end. Meanwhile, liquidity creation has a positive correlation with economic output and growth, as well as with business cycle fluctuations

(Berger and Sedunov, 2017; Davydov et al., 2021). Thus, the expansion of bank liquidity creation further contributes to the growth of the real economy.

Second, bank liquidity creation is not only affected by the liquidity provided by external monetary authorities, but also depends on the line of credit that banks are willing to provide (Bawazira et al., 2018). The development of fintech generates the effect of technology spillovers and improves the ability and willingness of banks to create liquidity. There is still a lack of diversity in the design, promotion and sales of traditional bank loan products. The credit business takes physical outlets as the main channel, and the channels into consumption and life scenes are relatively limited. The increasing maturity of digital technology is reducing technical barriers, and digital technology, originally monopolized by internet platforms, is gradually being applied by commercial banks. In terms of demand, the outbreak of COVID-19 has shifted a large number of economic activities from offline to online, which objectively promotes customer demand for online loans, wealth management, payment and settlement, and encourages banks to apply fintech to accelerate digital transformation. On the supply side, banks continue to improve their services and optimize their business models through the application of financial technology. An increasing number of banks are able to provide customers with online, personalized and diversified financial services, which not only expands banks' willingness to lend, but also effectively improves their efficiency and ability to serve the real economy. Therefore, banks use fintech to improve their ability to acquire customers, to promote the growth of illiquid assets such as loans, and thus to improve the level of bank liquidity creation. Therefore, related studies in the literature and the discussion above lead to our first hypothesis:

H1. The development of fintech increases bank liquidity creation.

The ability of fintech to influence liquidity creation varies across banks with different characteristics. First, in terms of bank size, compared with small banks, large banks have their own capital strength, have a superior layout of financial technology, and can make more sophisticated use of fintech in terms of product design, customer marketing, service innovation, and process management, so they have the ability to increase the line of credit, convert liquid funds into illiquid assets, create more liquidity for the real economy, and promote a higher level of economic output (Berger and Sedunov, 2017). In China, most of the large banks are state-owned, which have many branches, wide operating areas, and large business scale. At the same time, China has made the development of fintech a top strategy, which requires financial institutions to accelerate digital transformation and pay attention to technology-driven and data-driven innovation in finance.² State-owned banks have a special position and mission in the national economy, thus their business decisions tend to be more in line with government intentions, accelerating the implementation of fintech strategies and taking responsibility for providing liquidity to the real economy.

Second, in terms of ownership structure, the influence of shareholders on management tends to rise as ownership concentration increases (Shleifer and Vishny, 1986; Caprio and Levine, 2002), and liquidity creation is a way for bank shareholders to increase surplus (Berger and Bouwman, 2009). When banks use fintech to expand credit scale and improve liquidity creation, controlling shareholders have the motivation and power to increase risk-taking (Laeven and Levine, 2009) and improve liquidity creation (Yeddou and Pourroy, 2020). At the same time, the relatively high ownership concentration of banks is more convenient for management and decision-making. Facing the opportunities of fintech, this kind of bank can unify the shareholders' appeals, promote the management to make the decision of developing fintech rapidly, and improve the level of liquidity creation. Therefore,

compared to banks with lower ownership concentrations, banks with higher ownership concentrations are more sensitive to banking fintech.

Third, in terms of market power, banks with stronger market power could easily obtain monopoly profits to improve their risk resistance ability, and have strong ability to expand market share by virtue of higher market position and credit; meanwhile, it is easier for these banks to master advanced technology and obtain technological monopoly advantages. Under the enabling effect of fintech, it is easier for these banks to sink credit and form long-term stable lending relationships with counterparties (Petersen and Rajan, 1995), which tends to increase the willingness and scale of lending, resulting in generating more illiquid assets (Bawazira et al., 2018). However, banks with relatively weak market power have lower independent research and development capabilities in fintech and often need to follow banks with strong market power, so their ability to use fintech to issue credit and the level of liquidity creation are relatively weak. Thus, the above discussion leads to our second hypothesis:

H2. The impact of fintech on bank liquidity creation is more pronounced for banks with larger asset size, higher ownership concentration, and greater market power.

Moreover, liquidity creation and risk transfer are not equivalent. Fintech not only improves bank liquidity creation by changing the loan structure and expanding the retail business, but also increases bank risk-taking.

Specifically, banks specifically employ fintech in their lending operations, establish online customer acquisition channels, gather and analyze extensive volumes of data using advanced financial technologies. This approach helps reduce information asymmetry, effectively identify high-risk customers, and enable individuals who previously required collateral to secure loans based on their creditworthiness. Thus, banks reduce their reliance on collateral, then moderately reduce the mortgage guarantee loans and credit loans relative increase to absorb long-tail customers and increase the scale of credit loans to maximize profits, which will ultimately improve liquidity creation at the asset side. Meanwhile, banks use fintech to help more small and medium-sized customers avoid financial exclusion, especially by providing retail loans such as credit cards, online banking, and short-term loans to individuals or households, which can be profitable in the short term. Because it is more costly for banks to develop corporate customers, which requires due diligence on corporate collateral and cash flow evidence, and then to establish and maintain long-term bank-corporate relationships. Therefore, banks tend to greatly increase individual loans and relatively reduce corporate loans. However, banks will not greatly reduce corporate loans for long-term development, so liquidity creation and total liquidity creation on the asset side tend to increase eventually.

When banks absorb risk-free liquid deposits to issue riskier illiquid loans, a certain degree of risk transformation may result in greater liquidity creation (Berger and Bouwman, 2009), and excessive liquidity creation often leads to the financial crisis (Berger and Bouwman, 2017). On the one hand, fintech can increase banks' willingness to take risks. With the help of fintech, banks can increase online lending and change the structure of lending. Although fintech can reduce the degree of information asymmetry in the credit process and reduce the cost of loans (Demertzis et al., 2018; Buchak et al., 2018), the improvement of risk identification and management capabilities stimulates banks to improve their risk tolerance, which makes them overly rely on fintech risk management technology and increase risk-taking behavior. Due to the high threshold for commercial banks to apply fintech, banks need to invest a large amount of human, material and financial resources in the short term, experience a long technology incubation period, and the capital needs to be continuously followed up. In addition, banks need to continuously improve their services before new products form a market scale (Vernon, 1992). It greatly increases the R&D costs of commercial banks, therefore they have a strong incentive to invest in high-yield projects to balance the profits, thus increasing their risk-taking behavior.

² In January 2022, the People's Bank of China released the fintech development plan (2022–2025).

On the other hand, fintech accelerates bank credit expansion. Specifically, fintech can facilitate the expansion of banks' retail business, especially by helping banks search for customers more accurately and efficiently, providing convenient services, shortening the product development cycle, and offering more refined financial products. Then, retail operations are expanded with a focus on long-tail customers. Thus, banks provide more credit loans while reducing guarantee loans. The credit degree of bank customers is increasingly tilted to the group of SMEs and individuals, which are less qualified than large enterprises due to the lack of collateral. However, some studies show that the application of fintech still faces some problems in the specific implementation process of affecting the credit supply of SMEs (Liberti and Petersen, 2019). Although fintech uses non-traditional credit assessment methods, there is still a lack of sufficient evidence to validate their effectiveness in assessing default risk. A significant amount of data is not released during the data collection process, and some information is lost when 'soft' information is transformed into 'hard' information (Liberti and Petersen, 2019). Their credibility is increasingly difficult to verify, and the borrower's motivation is not properly captured (Jaksic and Marinc, 2019). Post-loan risk control relies more on regulators, and banks' fintech has no significant effect on reducing post-loan risk (Zhang et al., 2022). Especially after the outbreak of COVID-19, financial institutions ease credit access for new borrowers, thus increasing non-performing loans (Bao and Huang, 2021). It would also increase the level of bank risk-taking (Wang et al., 2021). The above discussion then leads to our third hypothesis:

H3. Fintech improves bank liquidity creation by changing credit structure, and increases bank risk-taking through the channel of liquidity creation.

3. Data and variables

We use a quarterly bank-level panel dataset encompassing 36 listed banks in China, spanning from Q1 2007 to Q2 2021. This dataset comprises 6 state-owned banks, 9 joint-venture banks, 13 urban banks, and 8 rural banks.³ By the end of Q2 2021, the total assets of the sample banks accounted for 65.18% of the total assets of China's banking sector,⁴ so the selected sample has relatively good representativeness. The majority of the data used in this study are collected from the WIND database, such as liquidity creation, bank balance sheet, and macroeconomic conditions. The bank fintech index is constructed based on the INFOBANK economic news database.

3.1. Measuring the development of bank fintech

We use word vector training and factor analysis techniques to perform semantic mining for bank news articles and construct the fintech development index. Since the information contained in the news can reflect economic phenomena (Askatas and Zimmermann, 2009), it is feasible to use this method to measure the development level of bank fintech. The construction process mainly includes three steps: constructing the basic lexicon of bank fintech, database selection and text mining, and index construction.

Step 1. Constructing the basic lexicon of bank fintech

Drawing upon relevant policies, literature, reports on fintech development of banks, field research, and expert insights, we have thoroughly organized and identified 155 keywords associated with bank fintech. These keywords are compiled and presented in Table 1 of online Appendix A.

³ These banks were chosen because they are listed banks and disclose data on a quarterly basis.

⁴ According to the People's Bank of China, the total assets of the banking industry in 2021Q2 were RMB 336 trillion yuan.

Table 1

Main variable definition.

Variable	Definition
<i>Bank liquidity creation</i>	
Asset liquidity creation (LCA)	The ratio of the difference between illiquid assets and liquid assets to total assets, multiplied by 1/2.
Total liquidity creation (LC)	Asset liquidity creation plus liquidity creation (see Berger and Bouwman, 2009).
<i>Bank fintech</i>	
Bank fintech index (BFI _{news})	The sum of keyword proportions calculated based on web news co-occurrence and text mining techniques.
Bank fintech index (BFI _{report})	The log value of the total number of types of keyword occurrences in the quarterly financial reports of each bank.
<i>Bank characteristics</i>	
Size	The natural log of bank total asset.
Profitability	The ratio of net income to total assets (ROA).
Non-interest income	Non-interest income calculated as the ratio of non-interest income to interest income.
Operating efficiency	The ratio of operating cost to operating income.
Book to market	The ratio of a bank's equity to book value.
<i>Macroeconomic conditions</i>	
GDP growth	Quarterly growth rate of Chinese real GDP.
Inflation	Quarterly growth rate of Chinese CPI.
M2 growth	Quarterly growth rate of Chinese M2.
Shibor	Quarterly average of the 7-day Shanghai Interbank Offered Rate.
Banking competition	Boone index is calculated to measure the competition in the banking industry (see Boone, 2008).

Note: This table presents the definitions of main variables used in this study.

Step 2. Database selection and text mining

First, we select the INFOBANK economic news database as our news corpus source for bank fintech. This data source includes news reports from websites of more than 1500 media around the world, as well as text information such as comprehensive newspapers and financial newspapers. The high coverage of China's economic and financial news of this database ensures that our bank fintech index well describes the real level of fintech development of the banking sector in China. Then, we select news reports from 2007Q1 (January 1, 2007) to 2021Q2 (June 2, 2021) in the China Economic News database of the INFOBANK database, filtering by the occurrence of banks. For each news text, we use Python web crawler technology from the *Selenium* library to simulate the browser behavior, capture all the news about banks, and construct the original news corpus of this study. Finally, we use the co-occurrence method to obtain keywords. Specifically, we use the "bank name + fintech keywords" method to obtain a news corpus, perform word segmentation and noise reduction, and count the total number of keywords for each fintech keyword k of bank i for each period t (month) $keyword_{kit}$ and the total number of word news reports about bank i in period t $Totalwords_{it}$.

Step 3. Index construction

First, we aggregate the occurrences of each fintech keyword to compute the total number of each fintech keyword of bank i in period t (each month) $\sum_k keyword_{kit}$. Second, we calculate the proportion of the total number of occurrences of all keywords of bank i to the total number of news words in period t as shown in Eq. (1) listed below, which is the bank fintech indicator of bank i in period t , measuring the development level of bank fintech or the attention to fintech.

$$BFI_{it} = \frac{\sum_k keyword_{kit}}{Totalwords_{it}} \quad (1)$$

In the following empirical analysis, the quarterly BFI is used in our regression analysis. Fig. 1 shows that China's bank fintech index has been on the rise, especially after the rise of internet finance in 2013. Some of the peaks in the shaded areas were the period when the Chinese government issued relevant policies and development plans to promote the development of fintech, and the index reached a new peak in 2020 after the COVID-19 outbreak. Fig. 1 illustrates the comparison of the

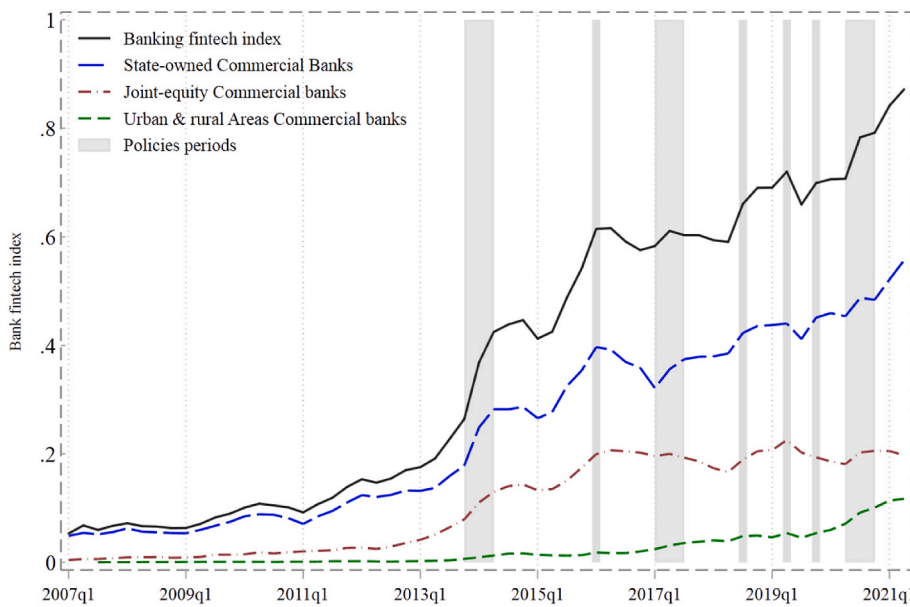


Fig. 1. Trends of bank fintech development in China. Note: This figure shows the trends in bank fintech development for different samples over time. The black line represents the general trend of fintech in the Chinese banking industry. The blue line is the fintech development level of the six state-owned commercial banks. The red line represents the fintech development level of the joint-venture commercial banks. The green line represents the fintech development level of urban and rural commercial banks. The grey shaded area indicates the periods during which the Chinese government has issued relevant policies to promote the development of bank fintech. (For interpretation of the references to colour in this figure legend, the reader is referred to the Web version of this article.)

bank fintech index across different categories, including state-owned banks, joint-venture banks, and urban and rural area banks. We can see that the level of fintech development or attention of state-owned banks is significantly higher than other banks, while urban and rural area banks have the lowest level. Joint-venture banks are in the middle. This is in line with the fact that state-owned banks and joint-venture banks are early and fast movers in fintech development.

In Fig. 2, we test the validity of the constructed bank fintech index. Specifically, we compare the quarterly frequency of the bank-level fintech index with the log value of the total number of quarterly patents of banks, and find that both are significantly increasing, with a correlation up to 81%. In addition, we compare the financial index with the Digital Finance Index of Qiu et al. (2018) and one of its components, the Digitalization Degree index developed by Peking University.⁵ These indices have been widely used by Chinese scholars, and our index is highly correlated with both, with correlation coefficients of 87% and 93%, respectively.

In addition, we also use the data from periodic reports, which truly reflect the operations of banks, to conduct text mining to construct indicators of the same fintech keyword lexicon shown in online Appendix A. To eliminate the influence of the length of periodic reports of different banks, we crawl keywords and count the total types of fintech keywords of each bank in the quarter to conduct a robustness test. Given that banks disclose little business information other than financial data in the first and third quarters, the frequency of the bank fintech index obtained from text analysis of annual reports is semi-annual.

3.2. Measuring bank liquidity creation

Liquidity creation, as defined by Diamond and Dybvig (1983), refers to the process wherein commercial banks employ liquid liabilities to facilitate investments in less liquid assets. Serving as a fundamental function of banks, liquidity creation is driven by the pursuit of profit

maximization and yields positive externalities for the real economy. However, it is important to note that an excessive level of liquidity creation can potentially accelerate the accumulation of bank risk.

In accordance with Berger and Bouwman (2009), we assess bank liquidity creation through a three-step process. First, we categorize balance sheet items into liquid, semi-liquid, and illiquid groups, considering factors such as duration and the ease of liquidation. Combined with the business model, accounting standards and macroeconomic environment of our Chinese banking system, we adjust the liquidity classification of some items. For example, residential mortgages are more difficult to securitize in China than in the US, so they are classified as illiquid rather than semi-liquid. As a result of the limited disclosure of off-balance sheet data by China's listed commercial banks, our calculation of bank liquidity creation is solely based on the information available on the balance sheet. The detailed breakdown of liquidity for balance sheet items can be found in online Appendix 2. In particular, deposits received on the liability side can be divided into time and savings deposits and demand and other customer deposits according to their maturities. However, since the financial data on WIND only releases the total amount of bank deposits, we collect the data of these two detailed deposits from the periodic financial reports of banks. Then they are divided into semi-liquid liabilities and current liabilities, which to some extent mitigates the problem that the liquidity creation measure of bank liabilities is too rough. However, the large number of missing values in the detailed deposit data reduces our total liquidity creation measure by almost one-third of the sample observations. Next, we assign different weights to the classified items. In accordance with the methodology outlined by Berger and Bouwman (2009), we assign specific weights to different asset and liability categories. As per their framework, the weight assigned to illiquid assets and liquid liabilities is 0.5, the weight assigned to illiquid liabilities and liquid assets is -0.5 , and the weight assigned to semi-liquid assets and semi-liquid liabilities is 0. Finally, the various items weighted above are added together to obtain the liquidity creation level of commercial banks. In order to make the liquidity creation indicators of banks of different sizes more comparable, we standardize the liquidity creation indicators with total assets as Eq. (2), Eq (3) and Eq (4):

$$LCA = \frac{0.5 \times (\text{Illiquid assets} - \text{liquid assets})}{\text{Total assets}} \quad (2)$$

⁵ The Digital Finance Index is compiled by the Financial Research Center of Peking University using Ant Financial household data, covering China's 31 provinces, 337 cities with more than 1700 county-level administrative divisions, and derive three architectural index, Pratt & Whitney financial coverage index, usage depth index and digital support degree index). Relevant sub-industries or business indexes are also included to comprehensively capture the development of digital inclusive finance.

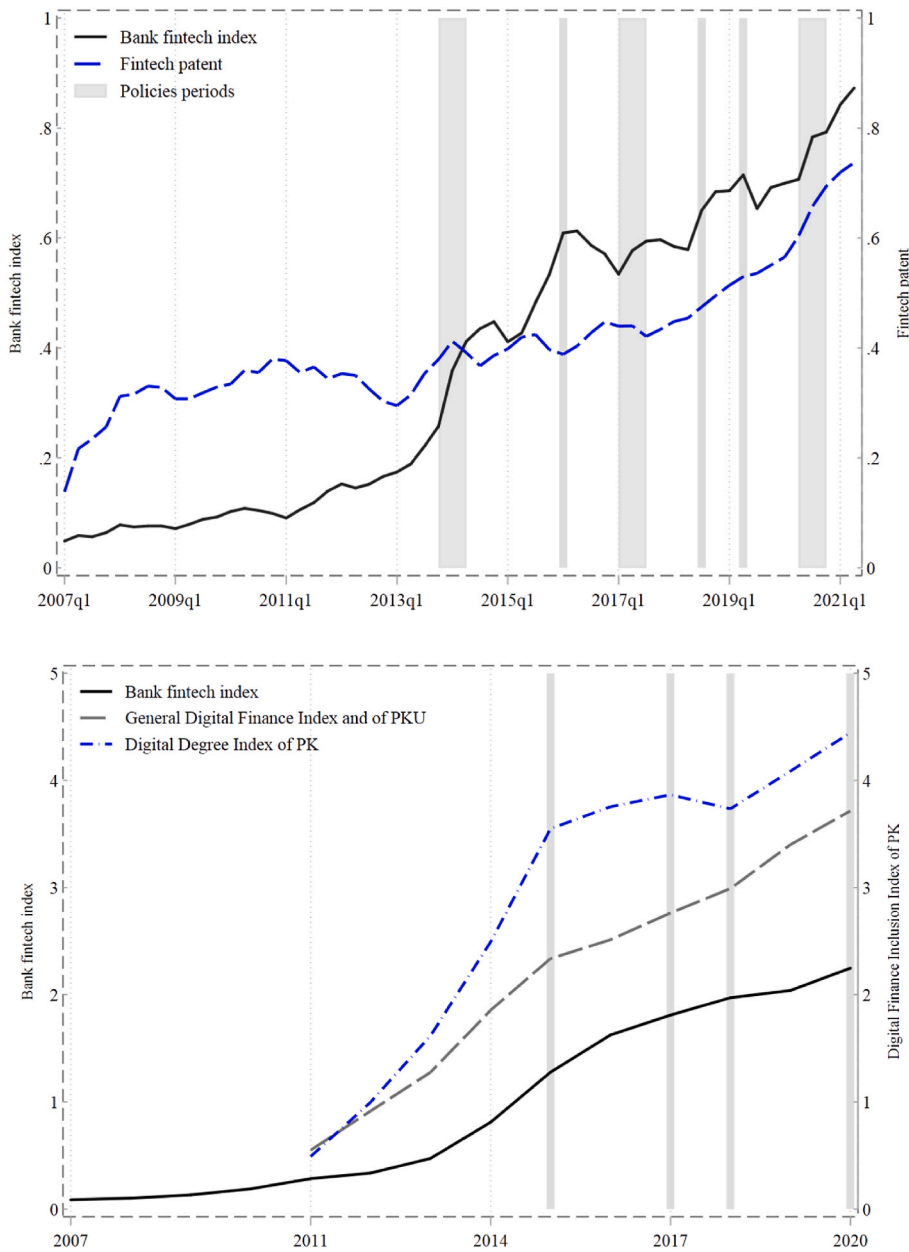


Fig. 2. Validation of bank fintech index

Note: This figure shows the evolution of the sample bank fintech over time. In the top panel, the black line is the quarterly fintech index of the banking industry constructed in this paper. The blue line is the total number of fintech patents after taking log values and unifying the dimensions in each quarter for the sample banks. In the figure below, the black line is the annual fintech index of the banking industry constructed in this paper. The blue and grey lines represent the Digital Finance Index and the Digitization Index developed by Peking University, respectively. These two indexes are obtained by a weighted sum of provincial GDPs. The grey shaded part indicates that China has issued relevant policies to vigorously promote the development of banking fintech. (For interpretation of the references to colour in this figure legend, the reader is referred to the Web version of this article.)

$$LCL = \frac{0.5 \times (\text{Liquid liabilities} - \text{Illiquid liabilities})}{\text{Total assets}} \quad (3)$$

$$LC = LCA + LCL \quad (4)$$

where LCA , LCL , and LC represent the liquidity creation of bank assets, the liquidity creation of bank liabilities, and the total liquidity creation of banks within the balance sheet, respectively. Finally, we obtain the quarterly bank liquidity creation indicators.

3.3. Control variables

Regarding the control variables, we select variables that capture both bank-specific characteristics and macroeconomic conditions, in accordance with the existing literature. In terms of bank characteristics, we include the following set of control variables: (1) Size is measured as the natural logarithm of total assets; (2) Profitability is evaluated using the return on assets (ROA), which is computed by dividing net income by total assets (Andreou et al., 2016; Cheng and Qu, 2020); (3) Noninterest

income proportion is calculated as the ratio of noninterest income to total operating revenue; (4) Operating efficiency is measured by the ratio of operating costs to operating income (Cheng and Qu, 2020); (5) Book to market is calculated as the ratio of stock market value to book value of equity, which reflects the growth potential and investment value of the bank in the capital market (DeYoung and Huang, 2021).

To account for the influence of the business cycle, we include five macroeconomic variables, as suggested by the literature (see Zhang et al., 2021; Zhao et al., 2022): (1) Macroeconomic development is assessed using the growth rate of real GDP, which serves as an indicator of the economy's position within the business cycle; (2) Inflation rate is measured by the growth rate of CPI, which is a representative indicator of macroeconomic imbalances. (3) Monetary policy, as measured by the growth rate of M2, can potentially affect the systemic risk of banks. A positive value indicates an expansionary policy, while a negative value implies a contractionary policy. (4) SHIBOR is the 7-day Shanghai Interbank Offered Rate. The higher the SHIBOR rate, the tighter the bank's liquidity. We also calculate the degree of competition in the banking industry, the BOONE index, following Boone (2008). The

higher the index, the less competitive the market. We interpolate some missing values of bank characteristics and winsorize all continuous variables at the 1% and 99% levels to exclude outliers.

3.4. Descriptive statistics

Table 2 reports the descriptive statistics of our main variables. For the main explanatory variable, within the interval of [0, 0.087], the mean value of the *BFI news* is 0.011, and the median is 0.009. Within the interval of [-0.125, 0.270], the mean value liquidity creation on assets (*LCA*) is 0.123, and the median is 0.139. The mean of total liquidity creation (*LC*) is 0.232, and the mean is 0.261, ranging from -0.089 and 0.420, respectively. This indicates that liquidity creation on assets is more volatile than liquidity creation.

4. Baseline model

Our baseline model is specified as follows:

$$Liquidity_Creation_{it} = \alpha + \beta BFI_{it-1} + \gamma Char_{it-1} + \theta Macro_t + f_i + \varepsilon_{it} \quad (5)$$

where the dependent variable *Liquidity_Creation_{it}* is the indicator of banks liquidity creation. *BFI_{it-1}* represents the index measuring the development of bank fintech. *Char_{it-1}* and *Macro_t* denote the vectors of bank characteristics and macroeconomic conditions, respectively. *f_i* is the time-invariance bank-specific effect, and ε_{it} is the error term. α , β , γ and θ are the coefficients to be estimated. To mitigate the problem of endogeneity, we include the one-year lagged bank fintech index and characteristics variables as control variables. We employ a fixed-effect estimator, which enables us to account for the correlations between the bank-specific variable and the other regressors.⁶ To control for potential heteroskedasticity and serial correlation, all regression coefficients are clustered and have robust standard errors.

4.1. Results of the baseline model

Table 3 presents the results of the baseline regression. The dependent variables used are asset liquidity creation and total liquidity creation. In columns (1) and (3), we include only bank fintech and bank

Table 2
Descriptive statistics for main variables.

	Mean	Median	SD	Max	Min	Obs
<i>Bank liquidity creation</i>						
<i>LCA</i>	0.123	0.139	0.081	0.27	-0.135	1112
<i>LC</i>	0.232	0.261	0.113	0.42	-0.089	776
<i>Bank fintech</i>						
<i>BFI news</i>	0.011	0.009	0.01	0.087	0	1112
<i>BFI report</i>	2.868	2.944	0.72	4.19	0.693	529
<i>Bank characteristics</i>						
<i>Size</i>	28.39	28.46	1.494	30.99	25.24	1112
<i>Profitability</i>	0.007	0.006	0.003	0.014	0.001	1112
<i>Non-interest income</i>	0.236	0.229	0.099	0.476	-0.024	1112
<i>Operating efficiency</i>	0.29	0.281	0.061	0.521	0.176	1112
<i>Book to market</i>	1.374	1.002	1.119	8.107	0.339	1112
<i>Macroeconomic conditions</i>						
<i>GDP</i>	0.075	0.07	0.039	0.183	-0.069	1112
<i>CPI</i>	0.026	0.022	0.018	0.08	-0.015	1112
<i>M2</i>	0.128	0.114	0.047	0.289	0.08	1112
<i>SHIBOR</i>	2.763	2.638	0.758	4.66	0.948	1112
<i>Banking competition</i>	-1.682	-1.538	2.636	4.686	-7.048	1112

Note: This table reports descriptive statistics for 36 listed banks in China over the period 2007Q1-2021Q2.

⁶ The fixed effects estimator is also preferable to the random effects estimator according to the Hausman test.

Table 3

The impacts of fintech on bank liquidity creation.

Dependent variable	(1)	(2)	(3)	(4)
	LCA	LCA	LC	LC
<i>Bank characteristics</i>				
<i>BFI news</i>	1.114** (2.33)	0.882** (2.20)	0.562 (1.04)	0.786* (1.74)
<i>Size</i>	-0.052*** (-2.87)	-0.051*** (-2.82)	-0.117*** (-5.31)	-0.109*** (-4.99)
<i>Profitability</i>	0.086 (0.09)	0.276 (0.32)	-0.297 (-0.29)	-0.88 (-0.84)
<i>Non-interest income</i>	0.206** (2.13)	0.132 (1.57)	0.239** (2.07)	0.164 (1.49)
<i>Operating efficiency</i>	0.062 (0.38)	0.070 (0.47)	0.126 (0.83)	0.131 (0.93)
<i>Book to market</i>	-0.008** (-2.34)	-0.025*** (-5.53)	-0.012* (-2.02)	-0.026*** (-4.23)
<i>Macroeconomic conditions</i>				
<i>GDP growth</i>		0.243*** (3.77)		0.192*** (2.91)
<i>Inflation</i>		1.211*** (4.35)		1.167*** (4.61)
<i>M2 growth</i>		-0.089 (-0.85)		0.094 (0.52)
<i>SHIBOR</i>		-0.032*** (-6.65)		-0.022*** (-4.54)
<i>Boone</i>		0.002*** (4.00)		0.003*** (4.74)
Constant	1.520*** (3.12)	1.603*** (3.19)	3.476*** (5.78)	3.318*** (5.46)
Bank fixed effect	YES	YES	YES	YES
Number of banks	1071	1071	757	757
Adjusted R ²	0.055	0.169	0.277	0.335

Note: This table reports the results for the impact of bank fintech on asset liquidity creation and total liquidity creation. We use heteroscedasticity-adjusted robust standard error clustering at the bank level. The *t* values are in parentheses. ***, **, and * denote significance at the 1%, 5%, and 10%, respectively.

characteristic variables as regressors, and in columns (2) and (4), we include macroeconomic conditions as additional control variables. The results show the significant positive correlation between bank fintech and asset liquidity creation as well as total liquidity creation,⁷ suggesting that an increase in bank liquidity creation is associated with the increasing level of bank fintech. Specifically, when the level of bank fintech increases by one standard deviation (0.010), bank liquidity creation on assets tends to increase by almost 0.88% ($\approx 0.882 \times 0.010$), and total bank liquidity creation on assets tends to increase by almost 0.79% ($\approx 0.786 \times 0.010$). Our results are in line with existing studies that show that the advancement of fintech contributes to the enhancement of liquidity creation by banks. Some studies have documented that the integration of technology and finance can help banks expand long-tail customers (Carletti, 2004; Yin et al., 2019; Zhang and Naceur, 2019), increase the scale of lending (Sheng, 2021), expand the scale of retail business, increase the supply of bank credit, increase the level of

⁷ In the baseline regressions, the coefficient on the impact of fintech on asset liquidity creation is larger than the coefficient on the impact on total liquidity creation. This may be related to the impact of bank fintech on liquidity creation on the liability side. The relationship between the two is tested and found to be negative. This may be related to the accounting method of the bank's liability side. Because banks use fintech to develop various applications embedded in financial products to absorb funds, and financial services such as outsourcing or commission financing are usually off-balance sheet items. In addition, many commercial banks in China have established wealth management subsidiaries to conduct wealth management business. Therefore, the financial funds of the bank are not included in the deposit account, but in the liabilities of the financial subsidiary, which ultimately affects the long-term capital investment of the parent company, thus reducing the liquidity creation of the debt side of the bank's balance sheet.

asset-side liquidity creation, and thus increase the total liquidity creation. Therefore, **H1** is supported.

The estimates of the control variables presented in **Table 3** further highlight the significance of these variables as determinants of bank liquidity creation. The regression results reveal a negative association between the size of banks and liquidity creation. This finding suggests that larger banks, given their prominent role in the financial system, are expected to provide a greater margin. The coefficients on the share of non-interest income are positive, which also indicates that the expansion of banks' non-interest income business increases illiquid assets and thus positively affects liquidity creation. The coefficients of book-to-market ratio are negative, indicating that the higher the market value of banks, the higher the risk of stock price decline, and thus the banks' ability to generate liquidity is weakened. *SHIBOR* has a negative coefficient, indicating that the higher the interbank market interest rate is, the less funds banks can borrow, thus limiting liquidity creation. The coefficient of Boone's index is positive, suggesting that the increase of banking monopoly can bring more liquidity.

4.2. Robustness tests

In this section, our objective is to strengthen the validity and reliability of the baseline regression results by conducting robustness tests from various perspectives.

4.2.1. Alternative measures of bank fintech

In this part, we employ alternative measures of bank fintech as key explanatory variables to conduct robustness tests. First, as mentioned above, we use Python to analyze the periodic reports of banks to obtain the keywords related to fintech to construct the bank fintech index by counting the number of types of fintech keywords in the annual and semi-annual reports of each bank. We add up the total number of types of keyword occurrences in the periodic reports of each sample bank in each period to construct each bank's fintech index (*BFI_Report*). In Panel A of **Table 4**, our results show that fintech significantly increases bank liquidity creation, which aligns with the results obtained from the baseline regression.

Second, following [Cheng and Qu \(2020\)](#), we choose Baidu search engine as our news data source, because it is the largest search engine in China. We use "bank name + fintech keyword" to crawl and obtain the quarterly amount of news for each keyword of each bank, and take the natural log. Then, we use principal component analysis to synthesize the fintech index (*BFI_pca*), which is used as an alternative proxy for the development of bank fintech. As shown in Panel B of **Table 4**, the bank fintech index constructed based on Baidu search engine is significantly positively correlated with bank liquidity creation, indicating that the baseline results are robust.

Third, we employ machine learning techniques to construct the bank fintech index. Specifically, we take "online, digital, fintech, big data, chain blocks, artificial intelligence, cloud computing, digital finance" as seed words. Based on the quarterly financial reports of banks, we use Word2Vec for training and expansion, and get 123 keywords related to the seed words (including seed words). Then, we calculate the word frequency of each quarterly financial report, and finally obtain the bank fintech index after standardization (*BFI_ml*). As indicated by the regression results in Panel C of **Table 4**, the fintech index constructed by the machine learning method is still positively correlated with the liquidity creation index.

Moreover, a higher level of bank fintech development is usually associated with more fintech patents. Therefore, we use the log value of the number of fintech patents as a proxy for the bank fintech development. The regression results are presented in Panel D of **Table 4**. Our findings demonstrate that the coefficients of the main explanatory variable (*BFI_patent*) remain significantly positive, providing further evidence for the robustness of the baseline regression results.

Finally, some studies on the Chinese market use the regional-level

Digital Finance Index developed by Peking University as a measure of fintech development ([Qiu et al., 2018](#)). The degree of digitalization index, one of its subcomponents, can reflect the level of financial digitalization in a region. We re-estimate the baseline regression on the fintech level of banks based on the digitalization index of the province where each bank is headquartered. In Panel E of **Table 4**, the regression coefficients are still significantly positive, confirming the robustness of the baseline regression.

Table 4
Robustness tests: Alternative measures of bank fintech.

Dependent variable	(1)	(2)
<i>Panel A: Using total number of types of bank fintech keywords crawled from bank annual reports.</i>		
	LCA	LC
<i>BFI_report</i>	0.045*** (4.38)	0.040*** (2.78)
Other variables	YES	YES
Bank fixed effect	YES	YES
Observations	518	513
Adjusted R ²	0.217	0.431
<i>Panel B: Using bank fintech Index constructed under principal component analysis based on number of banks fintech news via Baidu search engine.</i>		
	LCA	LC
<i>BFI_pca</i>	0.013*** (7.45)	0.010*** (7.16)
Other variables	YES	YES
Bank fixed effect	YES	YES
Observations	1080	767
Adjusted R ²	0.487	0.530
<i>Panel C: Using bank fintech Index constructed by text analysis of financial reports and machine learning techniques.</i>		
	LCA	LC
<i>BFI_ml</i>	0.110*** (6.92)	0.082*** (3.90)
Other variables	YES	YES
Bank fixed effect	YES	YES
Observations	1068	760
Adjusted R ²	0.213	0.354
<i>Panel D: Using bank fintech Index constructed based on bank patent data</i>		
	LCA	LC
<i>BFI_patent</i>	0.036*** (5.92)	0.043*** (5.00)
Other variables	YES	YES
Bank fixed effect	YES	YES
Observations	926	653
Adjusted R ²	0.247	0.372
<i>Panel E: Using the digitalization degree index of the city where the bank is headquartered</i>		
	LCA	LC
<i>BFI_PKU</i>	0.025** (2.15)	0.040*** (3.12)
Other variables	YES	YES
Bank fixed effect	YES	YES
Observations	858	633
Number of banks	0.316	0.228

Note: This table reports the results for the impact of fintech on bank liquidity creation using alternative measures of fintech. In Panel A, fintech development is proxied by the total number of types of bank fintech keywords crawled from bank annual reports. In Panel B, the bank fintech index is constructed using principal component analysis based on the number of bank fintech news items via the Baidu search engine. In Panel C, the bank fintech index is constructed using text analysis of financial reports and machine learning techniques. In Panel D, the fintech index is proxied by the number of bank patents. In Panel E, we use the Digitization Index developed by the Peking University to measure the degree of in the city where the bank is headquartered. For brevity, we report only the estimates of the coefficient on bank fintech. For brevity, we report only the estimates of the coefficient on bank fintech. All other regressors in the baseline model are also controlled. The *t* values of estimates are in parentheses. ***, **, and * denote significance at the 1%, 5%, and 10%, respectively.

4.2.2. Alternative measures of bank liquidity creation

As a robustness check, we consider alternative measures of bank liquidity creation. First, since the liquidity creation index used in the baseline regression is standardized by total assets, we adopt the sum of deposits and equity for standardization in this part. This reflects the extent to which each unit of funds (from depositors and shareholders) can create liquidity. As shown in Panel A of Table 5, the regression coefficients are positive, indicating that the development of bank fintech can bring positive liquidity creation.

Following DeYoung and Huang (2021), we classify liquidity creation into internal liquidity creation and external liquidity creation. Internal liquidity creation denotes the liquidity created by banks that engage in risky activities in order to generate more profits. External liquidity creation refers to the liquidity that banks provide to other economic agents after satisfying the requirements of maximizing their own profits or the private returns of their shareholders. From a policy perspective, external liquidity creation means that banks provide additional liquidity to the real economy. The liquidity created by banks circulates throughout the economy and generates positive social externalities. We

Table 5
Robustness tests: Alternative measures of bank liquidity creation.

Dependent variable	(1)	(2)
<i>Panel A: Using liquidity creation indicator calibrated by liquidity discount rate</i>		
	<i>LCA</i>	<i>LC</i>
<i>BFI_news</i>	1.335** (2.17)	1.220* (1.88)
Control variables	YES	YES
Bank fixed effect	YES	YES
Observations	1071	757
Adjusted R ²	0.162	0.242
<i>Panel B: Using internal liquidity creation indicators</i>		
	<i>in_LCA</i>	<i>in_LC</i>
<i>BFI_news</i>	0.194* (1.96)	0.307* (1.73)
Control variables	YES	YES
Bank fixed effect	YES	YES
Observations	1069	756
Adjusted R ²	0.301	0.525
<i>Panel C: Using external liquidity creation indicators</i>		
	<i>ex_LCA</i>	<i>ex_LC</i>
<i>BFI_news</i>	0.807* (1.84)	0.581 (1.08)
Control variables	YES	YES
Bank fixed effect	YES	YES
Observations	1069	756
Adjusted R ²	0.159	0.245

Note: This table reports the results for the impact of fintech on bank liquidity creation using alternative measures. In Panel A, we use a measure of liquidity creation standardized by total bank deposits and equity, rather than total assets. In Panel B, we use the bank internal liquidity creation indicator, which represents the liquidity created by banks to themselves when pursuing their own returns. In Panel C, we use the bank external liquidity creation indicator, which represents the externality to society measured by the residual of liquidity creation on the regression of private return proxies and the underlying variables that create liquidity. All other regressors in the baseline model are also controlled. The *t* values of estimates are in parentheses. ***, **, and * denote significance at the 1%, 5%, and 10%, respectively.

use the regression of Eq. (6) to decompose liquidity creation into its internal and external components:

$$Liquidity_Creation_{it} = k \cdot RE_{it} + \varepsilon_{it} \quad (6)$$

where $Liquidity_Creation_{it}$ represents the asset liquidity creation and total liquidity creation of bank *i* in quarter *t*, and RE_{it} is a series of indexes reflecting bank profits.⁸ The fitted value in Eq. (6) can be considered as the internal liquidity creation (In_LCA and In_LC), and the residual term can be considered as the external liquidity creation of bank (Ex_LCA and Ex_LC).

We re-estimate the baseline regressions for internal and external liquidity creation in Panels B and C of Table 5, respectively, and find that the application of fintech significantly promotes internal bank liquidity creation, while the impact on external liquidity creation on the asset side is also significantly positive, but the regression coefficients on total external liquidity are positive but insignificant. On the one hand, this may be due to the apparent lack of data on deposits with different maturities on the liabilities side when calculating total liquidity. On the other hand, banks' external liquidity creation includes not only liquidity creation to interbank, but also liquidity creation to real society. However, we find that the coefficients of bank fintech on interbank liquidity creation⁹ are negative but insignificant (see Table 1 in online Appendix C), indicating that banks use fintech more for retail business scenarios to support the real economy, rather than for interbank business, where idle funds circulate only in the banking industry.

To further investigate the impact of bank fintech development on the external liquidity created by banks for the real economy, we calculate the liquidity creation ($SLCA$ and SLC) for society as a whole using data disclosed by the People's Bank of China on the sources and uses of funds (RMB) by financial institutions. The asset liquidity creation and total liquidity creation is calculated by dividing the items of funds employed into liquid assets, semi-liquid assets and illiquid assets and assigning weights of $-1/2$, 0 and $1/2$ respectively, with reference to the liquidity creation structure of individual banks. Since the liquidity creation for society is calculated based on the source and use of funds in the whole banking industry, we calculate the fintech and each banking industry control variable by summing the individual bank fintech index and the bank-level control variable multiplied by the asset weight in each quarter. The asset weight is the share of the total assets of each bank in the total assets of the whole industry. Our empirical results show that fintech can significantly improve the liquidity creation for society and effectively serve the real economy (see Table 2 in online Appendix C).

4.2.3. Addressing endogeneity concerns

Although bank fintech may facilitate liquidity creation, the benefits of liquidity creation may also further develop fintech. And there might be some common factors affecting bank fintech and liquidity creation, such as policy, technology and human resources. Therefore, the model has endogeneity problems due to reverse causality and omitted variables.

We use multiple instrumental variables (IVs) and two-stage least squares (2SLS) to mitigate endogeneity. First, we select the attractive-

⁸ We use the one-year stock price returns predicted by the three-factor market model by considering more financial indicators, including return on equity (ROE) and return on assets (ROA). Revenue to cost ratio (inverse of cost to income ratio, *inverse_CRR*) market to book ratio (*MB*), annual change in market capitalization divided by total assets (ΔMVA), and net interest income divided by total assets (*NetIncoA*). We also incorporate bank's asset value (*SIZE*) and capital adequacy ratio (*CAR*), both of which are the basis of bank liquidity creation.

⁹ We construct interbank asset liquidity creation, liability liquidity creation, and aggregate liquidity creation indicators as follows: $Inter_LCA = \frac{0.5 \times (\text{Semiliquid assets} - \text{liquid assets})}{\text{Total assets}}$, $Inter_LCL = \frac{0.5 \times (\text{Liquid liabilities} - \text{Semiliquid liabilities})}{\text{Total assets}}$, $Inter_LC = Inter_LCA + Inter_LCL$

ness of fintech talents at the regional level as an instrumental variable, specifically the average salary of the financial industry in the province where each bank is headquartered, multiplied by the digital financial index of the province¹⁰. The salary level of the regional financial industry is an important factor in attracting financial talents (Cheng and Qu, 2020). At the same time, regions with a higher level of digital finance development are more likely to form the agglomeration effect of fintech talents due to the higher maturity of the fintech industry. In addition, labor migration is associated with the development of fintech (Arner et al., 2015). If the level of fintech development and the salary level of the financial industry in a certain region are more attractive, it will attract more fintech human capital, thus promoting the development of bank fintech. The fintech department of a bank is usually directly recruited and managed by the head office, and the head office pays more attention to the agglomeration of fintech talents in its region. Therefore, the attraction of fintech talents in the province where the bank's headquarters is located is highly correlated with the development of the bank fintech. The IV we select is exogenous because the attraction of fintech talents in the province where the bank headquarters is located reflects the competitiveness level of introducing fintech professionals in the region, and does not directly affect the operation decision of a single bank.

Second, we employ the provincial-level digital policy as an instrumental variable in our analysis. Specifically, we extract the word frequency of keywords related to the development of the digital economy in the annual government work reports issued by the provinces where the bank is headquartered.¹¹ Similarly, the attention paid to digital economy development policies in the provinces where the bank is headquartered reflects the extent to which the region promotes the application of digital emerging technologies in various industries, thereby facilitating the development of fintech in the banking sector; at the same time, bank headquarters usually actively respond to local policy calls, formulate long-term strategies and urge their branches to implement them in order to promote fintech development in the bank as a whole, while branches tend to implement headquarters instructions and do not tend to actively focus on policies. Therefore, it satisfies the conditions related to instrumental variables. The exogeneity of this IV is reflected in the fact that regional policies are external factors that do not directly influence the bank's traditional management decisions.

Third, we choose internet penetration as an instrumental variable.¹² The degree of internet penetration affects the development degree of bank fintech, but does not directly affect bank liquidity creation, so it also meets the requirements of exclusivity and relevance. In addition to the internet penetration rate, we choose two related to the development level of the internet and other emerging technologies as instrumental variables. The first one is the log value of the quarterly number of Internet broadband users. Its relevance is that the number of Internet users is huge, so the demand for financial services such as payment and settlement, credit and investment through the network is also increasing, which can promote the application of financial technology in various financial scenarios and promote the development of bank fintech. Exogeneity is manifested that the scale of Internet broadband users does not directly affect the operational decision of the bank's traditional business. The second one is the log value of the global market size of

cloud computing, big data, and artificial intelligence chip technology. The larger the scale of domestic and foreign underlying technology markets is, the more conducive for Chinese banks to develop financial technology. However, emerging technologies are exogenous factors and do not directly affect bank operations and liquidity creation.

The results of IV-2SLS are presented in Table 6. It can be seen that in the first stage, the instrumental variables constructed in this study and bank fintech are significantly positively correlated. In the second stage, the instrumental variables exhibit a significantly positive impact on liquidity creation, thereby confirming the robustness of the baseline regression results.

In addition, to allow for the persistence of bank liquidity creation, we transform Eq. (5) into a dynamic version by adding lags of the dependent variable as covariates and then using the dynamic generalized method of moments (GMM) estimator, which can also address potential endogeneity issues in our model (Blundell and Bond, 1998). As reported in Panel B of Table 7, we still find a significantly positive relationship between bank fintech and bank liquidity. Based on the above analysis, the results of our baseline model are robust.

5. Heterogeneity analysis

Based on the previous analysis, bank fintech may have distinct effects on liquidity creation for banks with various characteristics. To further investigate this heterogeneity, we conduct a comprehensive heterogeneity analysis by introducing three sets of dummy variables based on bank asset size, degree of ownership concentration, and market power. As shown in Eq. (7), these three sets of dummy variables are introduced into the model as interaction terms with bank fintech. The corresponding regression results are displayed in Table 8.

$$\begin{aligned} Liquidity_{Creation_{it}} = & \alpha + \gamma BFI_{it-1} + \beta BFI_{it-1} \times dummy_{it-1} \\ & + \lambda \cdot dummy_{it-1} + \eta Others + f_i + \varepsilon_{it} \end{aligned} \quad (7)$$

First, regarding bank size, we construct the dummy variable for bank size (*SCL_dummy*) according to the sum of each bank's capital and core liabilities as a proxy for scale equity and absorbed deposits.¹³ The size dummy variable is set to one for banks whose sum of capital and core liabilities is greater than or equal to the median, and zero for the rest. As stated in Panel A of Table 8, our analysis demonstrates a statistically significant positive relationship between bank fintech and liquidity creation. The interaction terms show that more advanced bank fintech significantly increases bank liquidity creation for banks with larger asset size. This could be attributed to the fact that banks with larger asset sizes typically have higher levels of capitalization. Therefore, they can facilitate the use of fintech for business transformation and product innovation, and have a strong lending capacity, thus creating more liquidity.

Second, with respect to the ownership structure of banks, we set the dummy variable (*Share_dummy*) equal to one for banks where the sum of the shareholdings of the ten largest shareholders is greater than or equal to the sample median, and equal to zero for the rest. The results for this heterogeneity test are reported in Panel B of Table 8. The coefficients on the interaction terms between bank fintech and the ownership structure dummy variable are found to be positive and statistically significant in relation to liquidity creation, which may suggest that the shareholders whose banks have a higher degree of bank fintech have more incentive to increase risk-taking and urge management to use fintech to expand credit to increase shareholder surplus. Meanwhile, higher ownership concentration also promotes efficient management and decision making of banks to develop fintech to create liquidity.

Third, the relationship between fintech and liquidity creation may vary among banks with different levels of market power. We set the

¹⁰ The average salary of the provincial financial industry is obtained from the CSMAR database and standardized $z_area_wage_{it} = \frac{area_wage_{it} - \min}{\max - \min}$. The Digital Finance Index uses the provincial digital finance total index developed by Peking University.

¹¹ The extracted keywords comprise the following: Internet, big data, artificial intelligence, blockchain, Internet of Things, cloud computing, digitalization, digital industry, 5G, digital infrastructure, Industrial digitalization, Cloud services, mobile payment, online, intelligence, wisdom, cloud, cloud platform, data security, data sharing, robotics, digital technology, digital economy.

¹² The data are obtained from the Wind database.

¹³ The scale of capital and core liabilities is proxied by the sum of total equity and absorbed deposits.

Table 6
Addressing endogeneity concerns: IV-2SLS.

Dependent variable	(1)	(2)	(3)	(4)
	Stage1	Stage2	Stage1	Stage2
<i>Panel A: Using Internet penetration</i>				
	<i>Internet penetration</i>	<i>LCA</i>	<i>Internet penetration</i>	<i>LC</i>
<i>BFI_news</i>	0.037*** (4.16)	25.981*** (4.50)	0.047*** (4.79)	23.849*** (4.76)
Control variables	YES	YES	YES	YES
Bank fixed effect	YES	YES	YES	YES
Observations	1035	1035	742	742
Kleibergen–Paap rk LM(P)	–	0.000	–	0.001
Cragg Donald Wald F	–	39.727	–	41.223
<i>Panel B: Using the Fintech talent attraction of the province where the bank is headquartered.</i>				
	<i>Fintech talent attraction</i>	<i>LCA</i>	<i>Fintech talent attraction</i>	<i>LC</i>
<i>BFI_news</i>	0.003*** (3.69)	13.127*** (2.77)	0.004*** (3.17)	15.515*** (2.58)
Control variables	YES	YES	YES	YES
Bank fixed effect	YES	YES	YES	YES
Observations	814	814	594	594
Kleibergen–Paap rk LM(P)	–	0.010	–	0.026
Cragg Donald Wald F	–	14.657	–	10.944
<i>Panel C: Using the degree of emphasis to digital economy development policies.</i>				
	<i>Policies</i>	<i>LCA</i>	<i>Policies</i>	<i>LC</i>
<i>BFI_news</i>	0.009*** (2.45)	23.809** (2.04)	0.010*** (2.70)	22.262** (2.34)
Control variables	YES	YES	YES	YES
Bank fixed effect	YES	YES	YES	YES
Observations	1065	1065	754	754
Kleibergen–Paap rk LM(P)	–	0.021	–	0.023
Cragg Donald Wald F	–	12.665	–	10.997
<i>Panel D: Using scale of Internet broadband users.</i>				
	<i>Internet broadband users</i>	<i>LCA</i>	<i>Internet broadband users</i>	<i>LC</i>
<i>BFI_news</i>	0.005*** (2.82)	45.092*** (3.01)	0.007*** (3.56)	38.606*** (3.60)
Control variables	YES	YES	YES	YES
Bank fixed effect	YES	YES	YES	YES
Observations	1071	1071	757	757
Kleibergen–Paap rk LM(P)	–	0.020	–	0.001
Cragg Donald Wald F	–	14.679	–	15.727
<i>Panel E: Using global scale of big data, cloud computing and chip technology.</i>				
	<i>global scale of technology</i>	<i>LCA</i>	<i>global scale of technology</i>	<i>LC</i>
<i>BFI_news</i>	0.013*** (6.44)	13.130*** (4.91)	0.011*** (3.26)	15.369*** (4.07)
Control variables	YES	YES	YES	YES
Bank fixed effect	YES	YES	YES	YES
Observations	1035	1035	743	743
Kleibergen–Paap rk LM(P)	–	0.001	–	0.005
Cragg Donald Wald F	–	49.758	–	26.813

Note: This table reports the results for addressing endogeneity concerns using IV-2SLS. From Panel A to Panel E, we select 5 instrumental variables each, namely, internet penetration, provincial level fintech talent attraction, provincial level digital economy development policy attention, internet broadband user scale, and global scale of big data, cloud computing and chip technology. We also apply the method of 2SLS instrumental variable technique. Kleibergen–Paap rk LM reports the p-value of the Kleibergen–Paap rank LM statistic for the under-identification test, Cragg Donald Wald F reports the F-statistic for rejecting the null hypothesis of weak instrumental variable. For the sake of brevity, we report only the estimates of the coefficient on bank fintech. All other regressors in the

baseline model are also controlled. The *t* values of estimates are in parentheses. ***, **, and * denote significance at the 1%, 5%, and 10%, respectively.

Table 7
Addressing endogeneity concerns: GMM estimator.

	LCA	LC
<i>BFI_news</i>	0.787* (2.00)	0.705*** (3.86)
L. dependent variable	0.953*** (26.35)	0.730*** (3.00)
Control variables	YES	YES
Observations	1053	465
AR-P (1)	0.000	0.033
AR-P (2)	0.404	0.042
Hansen J	23.55	3.788

Note: This table reports the results for addressing endogeneity concerns using the GMM estimator. AR(1) and AR(2) report the Arellano–Bond test for first and second order autocorrelation in the first-differenced errors. Hansen J reports the Hansen J statistic for the overidentification test. For the sake of brevity, we report only the estimates of the coefficient on bank fintech. All other regressors in the baseline model are also controlled. The *t* values of estimates are in parentheses. ***, **, and * denote significance at the 1%, 5%, and 10%, respectively.

Table 8
Heterogeneity analysis.

Dependent variable	(1)	(2)
	LCA	LC
<i>Panel A: Differences in asset size</i>		
<i>BFI_news</i> × <i>Scale_dummy</i>	3.194*** (2.89)	4.070*** (3.34)
<i>BFI_news</i>	−0.118 (−0.32)	−0.430 (−0.83)
Control variables	YES	YES
Bank fixed effect	YES	YES
Observations	1069	755
Adjusted R ²	0.265	0.392
<i>Panel B: Differences in ownership structure</i>		
<i>BFI_news</i> × <i>Share_dummy</i>	1.987** (2.04)	3.478*** (3.44)
<i>BFI_news</i>	0.169 (0.34)	−0.287 (−0.55)
Control variables	YES	YES
Bank fixed effect	YES	YES
Observations	1069	755
Adjusted R ²	0.215	0.385
<i>Panel C: Differences in market power</i>		
<i>BFI_news</i> × <i>Mono_dummy</i>	3.333*** (3.91)	4.036*** (3.70)
<i>BFI_news</i>	−1.513** (−2.23)	−2.084*** (−2.74)
Control variables	YES	YES
Bank fixed effect	YES	YES
Observations	1069	755
Adjusted R ²	0.270	0.406

Note: This table reports the heterogeneous impact of bank fintech on liquidity creation. In Panel A, we include the interaction terms of bank fintech and size dummy variable, which divides the sample into large and small banks. In Panel B, we include interaction terms of bank fintech and ownership concentration dummy variable, which divides the sample into different ownership structures. In Panel C, we include the interaction terms of bank fintech and market power dummy based on the Lerner index to reflect the degree of monopoly of banks. For the sake of brevity, we report only the estimates of the coefficients on bank fintech. All other regressors in the model are also controlled. The *t* values of estimates are in parentheses. ***, **, and * denote significance at the 1%, 5%, and 10%, respectively.

dummy variable (*mono_dummy*) equal to 1 if the banks have higher market power than the median of the sample banks, and 0 for the remaining banks. We measure the market power using the Lerner index following Beck et al. (2013). As reported in Panel C of Table 8, all interaction terms of the bank fintech and market power dummy variables are significantly positive. This indicates that banks with stronger market power or higher degree of monopoly are more likely to be the first to obtain the technological advantages brought by fintech, and are more able to use fintech to achieve business transformation. Meanwhile, banks with a strong market power are more inclined to expand their lending activities in order to attain monopoly profits, thereby increasing liquidity creation. Based on the above analysis, H2 is verified.

6. Further discussion

In this section, our objective is to investigate two specific questions. The first one is whether fintech increases bank liquidity creation by changing bank credit structure. The second one is whether fintech increases bank risk-taking by improving bank liquidity creation. Following Bertrand and Mullainathan (2001), we investigate above questions using a two-stage procedure. We denote the channel variables as *Channel* and set up the two-stage specification as follows. In the first stage, we run a regression similar to the baseline model to examine the impact of bank fintech on the channel variables:

$$\widehat{Channel}_{it} = \alpha_F + \gamma_F \cdot BFI_{it-1} + \eta_{it} \cdot Others + f_i + \varepsilon_{it} \quad (8)$$

where γ_F captures the influence of bank fintech on channel variables. This regression predicts the channel variable $\widehat{Channel}_{it}$ using only the information on the bank's fintech. In the second stage, we use the predicted value from the first stage as an explanatory variable to run a regression on the target dependent variables, namely liquidity creation or risk-taking:

$$Target - dependent - variables_{it} = \alpha + \kappa \cdot \widehat{Channel}_{it} + \eta_{it} \cdot Others + f_i + \varepsilon_{it} \quad (9)$$

where the estimated coefficient κ indicates how fintech affects bank liquidity creation and risk-taking through the channel variables. In the following sections, we use the above model to examine the channels through which fintech affects bank liquidity creation and risk-taking.

6.1. Bank fintech, loan structure, and liquidity creation

The development of bank fintech may change the credit structure (i.e., loan structure and customer structure) and thus affect liquidity creation. On the one hand, banks rely on fintech to obtain the "soft information" of borrowers for credit analysis, promoting the expansion of credit loans, while reducing the proportion of guaranteed loans relatively. On the other hand, as banks use fintech to expand their retail operations and focus more on lending to long-tail customers, especially individual customers. Thus, their proportion of personal loans is likely to rise, while the proportion of corporate loans is likely to fall relatively. Thus, we measure the channel variable loan structure using the proportion of credit loans in total loans and the proportion of guaranteed loans¹⁴ in total loans, respectively. In addition, we measure the customer structure using the proportion of personal loans and the proportion of corporate loans, respectively.

The results presented in Table 9 indicate that bank fintech is significantly positively related to credit loans and negatively correlated with guaranteed loans in the first-stage regression. This is probably because the development of bank fintech helps to collect and analyze

Table 9

The channel of fintech affecting bank liquidity creation: credit structure.

First-stage regression				
Dependent variable	(1)		(2)	
	Credit_loan		Guaranteed_loan	
BFI_news	0.246*	(1.71)	−0.600**	(−2.11)
Control variables	YES		YES	
Bank fixed effect	YES		YES	
Observations	1079		1079	
Adjusted R ²	0.222		0.328	
Second-stage regression				
Dependent variable	LCA		LC	
	Credit_loan		Guaranteed_loan	
$\widehat{Credit_loan}$	16.37***	(3.02)	13.50**	(2.24)
$\widehat{Guaranteed_loan}$		−2.241***		−1.847**
		(−3.02)		(−2.24)
Control variables	YES	YES	YES	YES
Bank fixed effect	YES	YES	YES	YES
Observations	1035	1035	743	743
Adjusted R ²	0.103	0.103	0.308	0.357

Note: This table reports the results of the test of whether fintech affects bank liquidity creation by changing the loan structure using a 2SLS method. In the first stage, we regress the bank fintech on the credit loan ratio and the guaranteed loan ratio, and predict the value of the channel factors using only the bank fintech information. In the second step, we regress the above predicted value on bank risk taking. *Credit_loan* is the share of credit loans of each bank. *Guaranteed_loan* is the proportion of guarantee loans of each bank. For brevity, we report only the estimates of the coefficient on bank fintech. All other regressors in the baseline model are also controlled. The *t* values of estimates are in parentheses. ***, **, and * denote significance at the 1%, 5%, and 10%, respectively.

information through more advanced technologies, reduce information asymmetry, identify more borrowers for loans, and thus reduce the dependency on collateral in the lending process. This allows many potential borrowers to rely on their credit information to obtain loans, resulting in a relative increase in the proportion of credit loans and a relative decrease in the proportion of guaranteed loans. In the second stage regression, the predicted value of credit loans due to bank fintech significantly positively affects bank liquidity creation, and the predicted value of guarantee loans due to bank fintech significantly negatively affects bank liquidity creation. The coefficient of the former is larger than that of the latter, indicating that bank fintech increases bank liquidity creation by expanding more credit loans. Although the reduction of part of the guarantee loans reduces the liquidity creation, the increased liquidity creation is greater than the reduced liquidity creation, so the total liquidity creation is increased. Thus, fintech increases bank liquidity creation by changing the credit structure and increasing credit lending.

The first-stage regression results presented in Table 10 reveal that bank fintech is significantly positive correlated with personal loans and significantly negative correlated with corporate loans. This is probably caused by banks expanding long-tail customer groups, such as micro, small, and medium customers, and increasing retail loans through fintech. As micro, small, and medium customers are more individual customers, the proportion of personal loans is relatively increasing while the proportion of corporate loans is relatively decreasing. In the second stage regression, the predicted value of personal loans using bank fintech has a significant and positive impact on bank liquidity creation, while the predicted value of corporate loans using bank fintech has a significant negative impact, but the coefficient of the former is larger than that of the latter. This suggests that bank fintech increases bank liquidity creation by expanding the share of personal loans, while the lower share of corporate loans somewhat reduces liquidity creation. The expansion of personal loans is larger than the reduction of corporate loans, so bank liquidity creation is generally increased. Therefore, the

¹⁴ Guaranteed loans include mortgage loans, pledge loans and third-party guarantee loans.

Table 10
The channel of fintech affecting bank liquidity creation: customer structure.

First-stage regression				
Dependent variable	(1)		(2)	
	Personal_loan		Corporate_loan	
BFI_news	1.050***		-0.784**	
	(3.58)		(-2.42)	
Control variables	YES		YES	
Bank fixed effect	YES		YES	
Observations	751		751	
Adjusted R ²	0.695		0.499	
Second-stage regression				
Dependent variable	LCA		LC	
	Individual_loan		Corporate_loan	
Individual_loan	14.18**		12.60*	
	(2.54)		(1.88)	
Corporate_loan		-1.941**		-1.725*
		(-2.54)		(-1.88)
Control variables	YES	YES	YES	YES
Bank fixed effect	YES	YES	YES	YES
Observations	726	726	473	473
Adjusted R ²	0.149	0.149	0.281	0.281

Note: This table reports the results of the test of whether fintech affects bank liquidity creation by changing the customer structure using a 2SLS method. In the first stage, we regress bank fintech on personal and corporate loans, predicting the value of the channel variable using only bank fintech information. In the second stage, we regress the predicted value on bank risk-taking. *Personal_loan* is the share of personal loans of each bank. *Corporate_loan* is the share of corporate loans of each bank. For brevity, we only report the estimates of the coefficients on bank fintech. All other regressors in the baseline model are also controlled. The *t* values of estimates are in parentheses. ***, **, and * denote significance at the 1%, 5%, and 10%, respectively.

development of fintech increases bank liquidity creation by changing the customer structure of loans.

We also examine why bank fintech can change the credit structure. Our results confirm that bank fintech can change the credit structure by increasing the loan size to micro, small, and medium customers and reducing the cost of credit. Detailed results are presented in online [Appendix D](#).

6.2. Bank fintech, liquidity creation, and risk-taking

As highlighted earlier, the progress of fintech can potentially result in an escalation of risk-taking behavior among banks. To test our hypothesis, we regress the fintech index on bank risk-taking variables. First, the ratio of risk-weighted assets to total assets (*RWA*)¹⁵ can reflect the willingness of bank to take risk as an *ex-ante* indicator of risk-taking. Second, we choose the non-performing loan ratio (*NPL*) as an *ex-post* risk-taking indicator, because the fintech cannot capture the real borrowing motivation of borrowers, and the credit issuance is without collateral, which may increase the non-performing loans. As reported in [Table 11](#), the coefficients of bank fintech on *RWA* and *NPL* are significantly positive, indicating that a higher level of bank fintech increases bank risk-taking.

We then examine the role of bank liquidity creation in the relationship between fintech development and bank risk-taking. We still use the two-stage regression approach introduced earlier to examine this, and the results are presented in [Table 12](#). In the first stage regression, we show that bank fintech is significantly positively correlated with bank liquidity creation, which is consistent with the results of the baseline regression. In the second stage regression, the predicted value of bank

¹⁵ Since there are some missing values of the scale of weighted risk assets of banks in the Wind database, we use the following formula to calculate the proportion of risk-weighted assets. $RWA = (\text{Total Equity} / \text{CAR}) / \text{Total Assets}$, where *CAR* is the capital adequacy ratio of the bank.

Table 11
The impact of fintech on bank risk-taking.

Dependent variable	(1)	(2)
	<i>RWA</i>	<i>NPL</i>
BFI_news	0.629*	0.043*
	(1.97)	(1.98)
Control variables	YES	YES
Bank fixed effect	YES	YES
Observations	1020	1116
Adjusted R ²	0.322	0.327

Note: This table reports the results for the impact of bank fintech on bank risk-taking. We regress the fintech indicator on the bank risk-taking indicator. *RWA* represents the ratio of risk-weighted assets to total assets. *NPL* is the non-performing loan ratio. All other regressors in the baseline model are also controlled for. For brevity, we only report the estimates of the coefficient on bank fintech. All other regressors in the baseline model are also controlled. The *t* value of estimates is in parentheses. The *t* values of estimates are in parentheses. ***, **, and * denote significance at the 1%, 5%, and 10%, respectively.

Table 12
The channel of fintech affecting bank risk-taking: liquidity creation.

First-stage regression				
Dependent variable	(1)	(2)		
	<i>LCA</i>	<i>LC</i>		
BFI_news	0.882**		0.786*	
	(2.20)		(1.74)	
Other variables	YES		YES	
Bank fixed effect	YES		YES	
Observations	1071		757	
Adjusted R ²	0.169		0.335	
Second-stage regression				
Dependent variable	<i>RWA</i>	<i>NPL</i>		
	<i>LCA</i>	<i>LC</i>		
<i>LCA</i>	0.878**		0.052**	
	(2.41)		(2.13)	
<i>LC</i>		0.884*		0.052*
		(1.90)		(1.84)
Control variables	YES	YES	YES	YES
Bank fixed effect	YES	YES	YES	YES
Observations	997	669	1071	757
Adjusted R ²	0.331	0.328	0.339	0.457

Note: This table reports the results of the test of whether fintech affects bank risk-taking through the liquidity creation channel using a 2SLS method. In the first stage, we regress bank fintech on liquidity creation and predict the value of the channel factors using only bank fintech information. In the second stage, we regress the predicted value on the bank risk-taking indicators. For brevity, we report only the estimates of the coefficient on bank fintech. All other regressors in the baseline model are also controlled. The *t* values of estimates are in parentheses. ***, **, and * denote significance at the 1%, 5%, and 10%, respectively.

liquidity creation is significantly positively associated with bank risk-taking. Therefore, [H3](#) is confirmed.

7. Conclusion

In this study, we analyze the influence of bank fintech on liquidity creation and risk-taking by utilizing a bank-level panel dataset comprising Chinese listed banks from 2007 to 2021. First, we employ text analysis to construct a new fintech index specifically tailored for the Chinese banking industry. Our index reveals a significant acceleration in fintech development within the Chinese banking sector during the sample period. Second, we measure bank liquidity creation from different perspectives. Our empirical findings consistently support our hypotheses, providing strong evidence that fintech significantly enhances the liquidity creation capabilities of Chinese listed banks. Third, our further analyses demonstrate that fintech affects bank liquidity creation by changing their credit structure (i.e., loan structure and

customer structure). Fourth, we examine the heterogeneous impact of fintech on bank liquidity creation and find that the positive effects are particularly pronounced for banks with larger asset size, higher ownership concentration, and greater market power. Finally, we show that fintech can amplify bank risk-taking through the channel of liquidity creation.

Based on the empirical results, this study offers several valuable policy implications. Firstly, it is crucial for Chinese banks to proactively adapt to the rapid development of fintech. They should clearly define the long-term strategic position of fintech-driven business development and accelerate their digital transformation efforts to enhance their ability to serve the real economy effectively. Additionally, banks with different characteristics should formulate their own fintech development plans and pathways. Specifically, small and medium-sized banks should leverage fintech to capitalize on their regional advantages and distinctive features. By strengthening the design of financial products, they can generate more liquidity for the real economy. Lastly, banks need to be mindful of the risks associated with the development of fintech in liquidity creation. Regulators should strengthen both micro- and macro-prudential supervision to prevent the accumulation of excessive risks. It is essential to fully harness the potential of bank fintech in facilitating liquidity creation while ensuring appropriate risk management measures are in place.

Declaration of competing interest

We hereby declare that there is no known competing financial interests or personal relationships that could have appeared to influence the work reported in the paper entitled “Bank Fintech, Liquidity Creation, and Risk-taking: Evidence from China”.

Data availability

Data will be made available on request.

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Appendix A. Supplementary data

Supplementary data to this article can be found online at <https://doi.org/10.1016/j.econmod.2023.106445>.

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